

MATH 455 ASSIGNMENT 4

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PROBLEM 1

Proof. The first observation is that after a translation we may assume without loss of generality that $f(x_0) = 0$. Certainly, this is a result of the linearity of the integral together with the fact that for each $n \in \mathbb{N}$ and $\gamma \in \mathbb{R}$ one has $\int_{-\pi}^{\pi} \gamma e^{-int} dt = 0$. Moreover, we may assume that $x_0 = 0$. To see this, we may pass to a translation $g(x) := f(x_0 + x)$. Now, we calculate the coefficients associated to this new function g at $x = 0$:

$$a_n^{(g)} = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(t) e^{-int} dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x_0 + t) e^{-int} dt = \frac{e^{inx_0}}{2\pi} \int_{x_0-\pi}^{x_0+\pi} f(u) e^{-inu} du$$

this last line is precisely the quantity $a_n^{(f)} e^{inx_0}$, periodicity. Therefore, if we can establish

$$\sum_{N \in \mathbb{Z}} a_n^{(g)} e^{in(0)} \sim g(0)$$

we will have since $g(0) = f(x_0)$ that $\sum_{N \in \mathbb{Z}} a_n^{(f)} e^{inx_0} \sim f(x_0)$ as we wish. We now carry this out, under the justified assumption that $x_0 = f(x_0) = 0$. For all $N \in \mathbb{Z}$ with $N \neq 0$:

$$\begin{aligned} \sum_{|n| \leq N} a_n e^{inx_0} &= \sum_{|n| \leq N} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt \right) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \sum_{|n| \leq N} e^{-int} dt \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} -\pi^\pi f(t) \left[\frac{e^{-i(N+1)t} - e^{iNt}}{e^{-it} - 1} \right] dt \end{aligned}$$

Of course, this corresponds to:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{f(t)}{e^{-it} - 1} \cdot e^{-i(N+1)t} dt - \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{f(t)}{e^{-it} - 1} \cdot e^{iNt} dt \quad (1)$$

To see that the Riemann-Lebesgue lemma applies to each of these terms, we use the differentiability of f at $x_0 = 0$. Certainly, by differentiability the limit

$$\lim_{t \rightarrow 0} \frac{f(t)}{e^{-it} - 1}$$

exists and is finite. Thence, this function $\frac{f(t)}{e^{-it} - 1}$ is bounded near $x_0 = 0$. By continuity over a compact set excluding a neighbourhood of x_0 , it follows that f is integrable on $[-\pi, \pi]$, whence we may apply this lemma to $\frac{f(t)}{e^{-it} - 1}$ and the result follows, since $f(x_0) = 0$ by assumption. $\square$

PROBLEM 2

Proof. We claim the operator $T : L^2([0, 1]) \rightarrow L^1([0, 1])$ given by $f \mapsto \int_0^1 f(y) dy$ is compact with $\sigma(T) = \{0\}$. We shall first show that this operator is bounded. To see this we write:

$$\|Tf\|_{L^2}^2 = \int_0^1 |Tf(x)|_2^2 dx \leq \int_0^1 \int_0^x |f(y)| dy dx \leq \int_0^1 \|f\|_{L^1} dx$$

yielding $\|Tf\|_{L^2}^2 = \|f\|_{L^1([0,1])}^2$. Recall now from the previous assignment that if f is supported on a set of finite measure E and L^2 then $\|f\|_{L^1} \leq \sqrt{\mu(E)} \|f\|_{L^2}$. Applying this above we obtain $\|Tf\|_{L^2}^2 \leq \|f\|_{L^2}^2$ proving boundedness.

By a result from Stein it suffices to prove that the Volterra operator is Hilbert-Schmidt. To see this, we note that we may always write:

$$Tf(x) = \int_0^x f(y) dy = \int_{[0,1]} K(x,y) f(y) dy, \quad K(x,y) := \begin{cases} 1 & y \leq x \\ 0 & \text{else} \end{cases} \quad (2)$$

Of course, $\int_{[0,1]^2} K(x,y) dx dy = \int_0^1 \int_0^1 K(x,y) dy dx = \int_0^1 \int_0^x dy dx = \int_0^1 x dx < \infty$. Proving that this linear operator is indeed Hilbert-Schmidt. It is now obvious that $\dim(L^2([0,1])) = \infty$ and therefore we must always have $0 \in \sigma(T)$. To see the reverse inclusion, assume instead there is some $\lambda \neq 0$ in the spectrum of T . Then:

$$Tf(x) = \lambda f(x) \text{ a.e., } f \in L^2([0,1]),$$

Hence, $\int_0^x f(y) dy = \lambda f(x)$. We now claim that f is in-fact continuous. To see this assume wlog that $y < x$ and write

$$|f(x) - f(y)| \leq \frac{1}{|\lambda|} \int_y^x |f(\alpha)| d\alpha = \frac{1}{|\lambda|} \int_0^1 |f(\alpha)| \mathbb{1}_{[y,x]} d\alpha = \frac{1}{|\lambda|} \langle |f|, \mathbb{1}_{[x,y]} \rangle$$

It then follows from Cauchy-Schwarz that $|f(x) - f(y)| \leq C \|f\|_{L^2} \sqrt{x-y}$, establishing the continuity of f . Therefore, f is in particular Riemann integrable. Interpreting the equality $\int_0^x f(y) dy = \lambda f(x)$ as a Riemann integrable, we find that f is differentiable on $[0,1]$. Thus yielding,

$$f(x) = \lambda f'(x) \implies f(x) = C_1 e^{x/\lambda} \quad (3)$$

This implies finally that:

$$\int_0^x f(y) dy = C_1 \int_0^x e^{y/\lambda} dy = C_1 \lambda (e^{x/\lambda} - 1) = \lambda C_1 e^{x/\lambda}$$

implying that $C_1 = 0$ and therefore that $f = 0$: a contradiction. $\square$

EXERCISE 1

Proof.

- (1) Integrable functions are finite almost-everywhere, and therefore this will follow immediately from (2).
- (2) We now calculate this norm:

$$\begin{aligned} \|f * k\|_{L^2}^2 &= \int_{\mathbb{R}^d} \left| \int_{\mathbb{R}^d} f(x-y) k(y) dy \right|^2 dx \\ &\leq \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |f(x-y)| |k(y)|^{1/2} \cdot |k(y)|^{1/2} dy \right)^2 dx \end{aligned}$$

By Cauchy-Schwarz we find that (since $|k|^{1/2}$ is $L^2(\mathbb{R}^d)$):

$$\begin{aligned}
\|f * k\|_{L^2}^2 &\leq \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |f(x-y)|^2 |k(y)| dy \cdot \int_{\mathbb{R}^d} |k(y)| dy \right) dx \\
&= \|k\|_{L^1} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f(x-y)|^2 |k(y)| dx dy \\
&= \|k\|_{L^1} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f(x-y)|^2 |k(y)| dx dy \\
&= \|k\|_{L^1} \int_{\mathbb{R}^d} |k(y)| \int_{\mathbb{R}^d} |f(x)|^2 dx dy \\
&= \|k\|_{L^1}^2 \|f\|_{L^2}^2
\end{aligned}$$

taking square-roots concludes the proof. Let us observe that (1) is now certainly satisfied.

- (3) We know that this formula holds true for $f \in L^1(\mathbb{R}^d)$ by our assignment(s) in Analysis III, yet we prove it here both for a review and completeness of this problem.

Case $f \in L^1(\mathbb{R}^d)$: This is “computational”:

$$\begin{aligned}
\widehat{f * k}(\xi) &= \int_{\mathbb{R}^d} e^{-2\pi i x \cdot \xi} (f * k)(x) dx \\
&= \int_{\mathbb{R}^d} e^{-2\pi i x \cdot \xi} \left(\int_{\mathbb{R}^d} f(x-y) k(y) dy \right) dx \\
&= \int_{\mathbb{R}^d \times \mathbb{R}^d} e^{-2\pi i x \cdot \xi} f(x-y) k(y) dx dy \\
&= \int_{\mathbb{R}^d \times \mathbb{R}^d} f(x-y) e^{-2\pi i (x-y) \cdot \xi} \cdot k(y) e^{-2\pi i y \cdot \xi} dx dy
\end{aligned}$$

Now, using Fubini we recover by translation-invariance:

$$\widehat{f * k}(\xi) = \int_{\mathbb{R}^d} k(y) e^{-2\pi i y \cdot \xi} \int_{\mathbb{R}^d} f(x) e^{-2\pi i x \cdot \xi} dx dy = \widehat{f}(\xi) \cdot \widehat{k}(\xi)$$

Case $f \in L^2(\mathbb{R}^d)$: It is clear that $\mathfrak{Q} := L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$ is dense in $L^2(\mathbb{R}^d)$. Let now $(f_n) \subseteq \mathfrak{Q}$ be a sequence of functions converging to f in $L^1(\mathbb{R}^d)$. Therefore, we have by linearity of the Fourier transform together with Plancherel’s identity:

$$\|\widehat{f_n * k} - \widehat{f * k}\|_{L^2} = \|f * k - f_n * k\|_{L^2} = \|(f - f_n) * k\|_{L^2} \stackrel{(b)}{\leq} \|k\|_{L^1} \cdot \|f_n - f\|_{L^2}$$

It is ultimately a consequence of the Borel-Cantelli lemma together with completeness that in any $L^p(\mathbb{R}^d)$ space given $\lim_{n \rightarrow \infty} \|f_n - f\|_{L^p} = 0$ we may extract a subsequence (f_{n_k}) converging to f almost everywhere. Let now (f_{n_k}) be such a subsequence in $\mathfrak{Q}$. Now, for almost all ξ we have:

$$\widehat{(f * k)}(\xi) = \lim_{n_k \rightarrow \infty} \widehat{(f_{n_k} * k)}(\xi) = \lim_{n_k \rightarrow \infty} \widehat{k}(\xi) \cdot \widehat{f_{n_k}}(\xi) = \lim_{n_k \rightarrow \infty} \widehat{k}(\xi) \cdot \widehat{f_{n_k}}(\xi) \quad (4)$$

Again, by Plancherel’s identity we have $\|\widehat{f_n} - \widehat{f}\|_2 = \|f_n - f\|_2$ and therefore we may pass to another subsequence to have $\widehat{f_{n_k}}(\xi) \rightarrow \widehat{f}(\xi)$ almost-everywhere. This gives us the result in the above.

- (4) This is an immediate consequence of the proof in (c).

□

EXERCISE 2

We first construct a simple bijection $\gamma : L^2(\mathbb{R}_+, dt/t) \rightarrow L^2(\mathbb{R})$ where we send $f(t) \mapsto g(x) := f(e^{-x})$ for $x \in \mathbb{R}$. Moreover, let us consider the inverse map:

$$\tilde{\gamma} : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R}_+, dt/t), \quad g(x) \mapsto f(t) = g(-\ln t)$$

It is obvious that $\gamma \circ \tilde{\gamma} = \tilde{\gamma} \circ \gamma = \mathbf{1}(\cdot)$ and hence we find that γ is a bijection. To see that this is well defined, we calculate for $f \in L^2(\mathbb{R}_+, dt/t)$:

$$\|\gamma(f)\|^2 = \|g\|^2 = \int_{-\infty}^{\infty} |f(e^{-x})|^2 dx \xrightarrow{t=e^{-x}} \int_0^{\infty} \frac{|f(t)|^2}{t} dt = \|f\|^2$$

We now claim that in fact, the map $\mathcal{M}$ is simply the composition $\mathcal{F} \circ \gamma$. To see this, we compute for $f \in L^2(\mathbb{R}_+, dt/t)$:

$$\mathcal{F}(\gamma(f))(\xi) = \int_{\mathbb{R}} f(e^{-x}) e^{-ix\xi} dx \xrightarrow{t=e^{-x}} \int_{\mathbb{R}_+} \frac{f(t)}{t} t^{i\xi} dt = (\mathcal{M}(f))(\xi)$$

Hence, this transform is bijective (as it is the composition of two bijections). To see that it “preserves the norm” (up to a desired constant), we note that by what we have determined above one has by Plancherel’s identity (for a slightly different definition of the Fourier transform, see the course notes rather than Stein)

$$\|\mathcal{M}(f)\|^2 = \|\mathcal{F}(\gamma(f))\|^2 = 2\pi \int_{-\infty}^{\infty} |f(e^{-x})|^2 dx = 2\pi \int_{\mathbb{R}_+} \frac{|f(t)|^2}{t} dt$$

which concludes the proof.

EXERCISE 6

Assume that $\Omega \subseteq \mathbb{C}$ is a non-empty open set. We must show that $\mathcal{H}(\Omega)$ is closed in $L^2(\Omega)$. Let (f_n) be a Cauchy sequence in $\mathcal{H}(\Omega)$: it is also Cauchy in $L^2(\Omega)$ and therefore has a limit $f \in L^2(\Omega)$; where $\|f_n - f\|_{L^2(\Omega)} \rightarrow 0$ as $n \rightarrow \infty$. Now fix a compact set $\Lambda \subset \Omega$. Let $z \in \Lambda$ be given: since Λ is compact and Ω^C is closed we have $\delta = d(\Lambda, \Omega^C) > 0$. Now for each $z \in \Lambda$ we consider:

$$f(z) = \frac{1}{\pi\delta^2} \iint_{B(z,\delta)} f(\zeta) d\zeta \tag{5}$$

To see the equation above, recall that the *mean-value property* holds for harmonic functions, and that the Cauchy-Riemann equations imply that both the real and imaginary parts of a holomorphic function are harmonic in the domain. Therefore, the above yields:

$$\begin{aligned} |(f_n(z) - f(z))^2| &= |f_n(z) - f(z)|^2 = \frac{1}{\pi\delta^2} \left| \iint_{B(z,\delta)} (f_n(\zeta) - f(\zeta))^2 d\zeta \right| \\ &\leq \frac{1}{\pi\delta^2} \iint_{B(z,\delta)} |f_n(\zeta) - f(\zeta)|^2 d\zeta \\ &\leq \frac{1}{\pi\delta^2} \|f_n - f\|_{L^2(\Omega)}^2 \end{aligned} \tag{*}$$

Where we have used that sums and natural powers of holomorphic functions are again holomorphic. Then, since $\|f_n - f\|_{L^2(\Omega)} \rightarrow 0$ and $z \in \Lambda$ was chosen arbitrarily the inequality above in $(*)$ proves that $\|f_n - f\|_{\infty} \rightarrow 0$ as $n \rightarrow \infty$. That is, $f_n \rightarrow f$ uniformly on every compact subset of Ω . A well known consequence of Morera’s theorem guarantees that f is then holomorphic on Ω .

EXERCISE 14

After a moments reflection, we note that it is much better to begin with the implication ($\Leftarrow$). Suppose now that F is absolutely continuous and that $F'(x) = G(x)$ for almost all x . We wish to show that G is the weak derivative of F . That is, we must show that for each smooth function $\varphi : [a, b] \rightarrow \mathbb{R}$, with φ smooth on (a, b) and $\varphi(a) = \varphi(b) = 0$ one has the subsequent identity:

$$\int_a^b F(x)\varphi'(x)dx = - \int_a^b G(x)\varphi(x)dx, \quad \forall \varphi \quad (6)$$

Now, it follows that φ is smooth, and is especially absolutely continuous on $[a, b]$. Then, the product $(F\varphi) := F(x)\varphi(x)$ is also absolutely continuous on $[a, b]$, since $[a, b]$ is a bounded interval. From our usual theory of absolutely continuous functions, this function $(F\varphi)$ is differentiable almost-everywhere and satisfies the fundamental theorem of calculus. Now, we note that since $\varphi(a) = \varphi(b) = 0$ we have immediately that

$$\int_a^b \frac{d(F\varphi)}{dx}(t) dt = (F\varphi)(t)|_{t=a}^{t=b} = F(b)\varphi(b) - F(a)\varphi(a) = 0$$

On the other-hand, we find that

$$\frac{d(F\varphi)}{dx} = F'(x)\varphi(x) + F(x)\varphi'(x)$$

That is,

$$0 = \int_a^b (F'(x)\varphi(x) + F(x)\varphi'(x)) dx = \int_a^b (G(x)\varphi(x) + F(x)\varphi'(x)) dx$$

where we have used that $F' = G$ almost-everywhere.

($\Rightarrow$). Assume that G is the weak-derivative of F on $[a, b]$. That is, for all φ as above we have:

$$\int_a^b F(x)\varphi'(x)dx = - \int_a^b G(x)\varphi(x)dx$$

Now, consider the function $H(x) := \int_a^x G(t)dt$ defined on $[a, b]$. This is clearly an absolutely continuous function. Hence, it is differentiable almost-everywhere and we get $H'(x) = G(x)$ almost everywhere. We may repeat the argument above to show that G is then the **weak derivative** of $H(x)$ (this is why we started with the above implication!). It now follows that;

$$\int_a^b F(x)\varphi'(x)dx = - \int_a^b G(x)\varphi(x)dx = \int_a^b H(x)\varphi'(x)dx$$

Therefore, $\int_a^b [F - H](x)\varphi'(x)dx = 0$ for all φ . This implies by their density that $F = H$ almost-everywhere. Now, since H was taken to be absolutely continuous the proof is complete.

EXERCISE 15

We begin with a brief calculation. We may define a differential operator $L := \left(\frac{\partial}{\partial x}\right)^\alpha$. Now, the adjoint-operator denoted L^* is given by

$$L^* = (-1)^{|\alpha|} \left(\frac{\partial}{\partial x}\right)^\alpha \quad (7)$$

Whence, it follows from well-known properties of $\mathcal{F}[\cdot] = \widehat{\cdot}$ that for all $\varphi \in C_0^\infty(\mathbb{R}^d)$.

$$\overline{\mathcal{F}[L^*\varphi](\xi)} = \overline{\mathcal{F}\left[(-1)^{|\alpha|}\left(\frac{\partial}{\partial x}\right)^\alpha \varphi\right](\xi)} = \overline{(-1)^{|\alpha|}(2\pi i\xi)^\alpha \mathcal{F}[\varphi](\xi)} \quad (8)$$

$$= (2\pi i\xi)^\alpha \overline{\mathcal{F}[\varphi](\xi)} \quad (9)$$

Now we show ($\implies$). Assume that $g = Lf$ weakly. Then, for all $\varphi \in C_0^\infty(\mathbb{R}^d)$ we have $\langle g, \varphi \rangle = \langle f, L^*\varphi \rangle$. Then, for each $\varphi \in C_0^\infty$ we have by Plancherel

$$\begin{aligned} \langle \widehat{g}, \widehat{\varphi} \rangle &= \langle g, \varphi \rangle = \langle f, L^*\varphi \rangle = \langle \widehat{f}, \widehat{L^*\varphi} \rangle = \int_{\mathbb{R}^d} \mathcal{F}[f](\xi) \overline{\mathcal{F}[L^*\varphi](\xi)} d\xi \\ &= \int_{\mathbb{R}^d} \widehat{f}(\xi) (2\pi i\xi)^\alpha \overline{\mathcal{F}[\varphi](\xi)} d\xi \\ &= \langle (2\pi i\xi)^\alpha \widehat{f}, \widehat{\varphi} \rangle \end{aligned}$$

which concludes the proof since we can choose any $\varphi \in C_0^\infty$, which is dense (and this equality follows from Plancherel).

($\impliedby$) We now assume that $\widehat{g}(\xi) = (2\pi i\xi)^\alpha \widehat{f}(\xi)$ a.e. We show that for $\varphi \in C_0^\infty(\mathbb{R}^d)$ we have $\langle g, \varphi \rangle = \langle f, L^*\varphi \rangle$. To see this, let us write out:

$$\begin{aligned} \langle g, \varphi \rangle &= \int_{\mathbb{R}^d} g(x) \overline{\varphi(x)} dx = \int_{\mathbb{R}^d} \mathcal{F}[g](\xi) \cdot \overline{\mathcal{F}[\varphi](\xi)} d\xi = \int_{\mathbb{R}^d} \mathcal{F}[f](\xi) \cdot (2\pi i\xi)^\alpha \overline{\mathcal{F}[\varphi](\xi)} d\xi \\ &= \int_{\mathbb{R}^d} \mathcal{F}[f](\xi) \cdot \overline{\mathcal{F}[L^*\varphi](\xi)} d\xi = \langle \widehat{f}, \widehat{L^*\varphi} \rangle \end{aligned}$$

where we have used the equality in (9) together with repeated use of Plancherel. Since this last term is exactly $\langle f, L^*\varphi \rangle$ the proof is complete.

EXERCISE 16

We heed the hint, and show that $\widehat{f}(\xi) = \mathcal{F}[f](\xi) \in L^1(\mathbb{R}^d)$. Certainly, we calculate

$$\begin{aligned} \int_{\mathbb{R}^d} |\widehat{f}(\xi)| d\xi &= \int_{\mathbb{R}^d} |\widehat{f}(\xi)| |1 + |\xi|^2|^{\frac{\alpha}{2}} |\widehat{f}(\xi)| |1 + |\xi|^2|^{-\frac{\alpha}{2}} d\xi \\ &\leq \left(\int_{\mathbb{R}^d} |\widehat{f}(\xi)|^2 |1 + |\xi|^2|^\alpha d\xi \right)^{\frac{1}{2}} \cdot \left\| |1 + |\xi|^2|^{-\frac{\alpha}{2}} \right\|_{L^2} \end{aligned}$$

Whence,

$$\|\widehat{f}\|_{L^1} \leq C_1 \left(\int_{\mathbb{R}^d} |\widehat{f}(\xi)|^2 |1 + |\xi|^2|^\alpha d\xi \right)^{\frac{1}{2}}$$

for some $C_1 > 0$ large. Therefore,

$$\|\widehat{f}\|_{L^1}^2 \leq C_2 \int_{\mathbb{R}^d} |\widehat{f}|^2 d\xi + C_2 \int_{\mathbb{R}^d} |\xi^\alpha \widehat{f}(\xi)|^2 d\xi$$

We invoke Plancherel's identity to deduce that the first quantity is simply $\|f\|_{L_2}^2$ and the last term is equal to $C_3 \left\| \frac{\partial^\alpha}{\partial x^\alpha} f \right\|_{L_2}^2$ by a class result together with Plancherel's identity. Therefore, $\widehat{f} \in L^1(\mathbb{R}^d)$.

We now pass to the fourier inverse transform, to find that

$$f(x) = \int_{\mathbb{R}^d} \widehat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi$$

where the right-hand side is clearly a continuous function of x (dominated convergence theorem). To see that the right-hand side of this expression is bounded, we use continuity and the Riemann Lebesgue lemma to observe that the RHS tends to 0 as $x \rightarrow \infty$.